

Determining latitude differential rotational speed of the sun by via tracking a sunspot

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1 Introduction

Due to the majority of the sun consisting of plasma, determining rotational speed across the photosphere can prove to be challenging, but with the use of a solar telescope and key software the speed near the solar equator can be determined. This technique could perhaps then be applied to other lines of latitude.

The sun forms the centre of our solar system, and is one of the most studied objects in all of human existence. From mythical interpretations in religions (such as Ra in ancient Egyptian mythology), to being studied by various physicists in the more recent ages (such as observing the transit of Venus in 1769 to calculate the astronomical unit (AU) [3].)

One of the qualities that the sun has, alongside many other celestial bodies within the solar system, is that it rotates around its own axis (which has a tilt of around 7.25 degrees [5]). The sun is also composed of several layers (similar to Earth's inner core, outer core, mantle and crust). The sun has the following layers [1]

- The Core
- The Radiative zone
- The Convection zone
- Photosphere, the deepest layer that can be observed and studied (the other layers within cannot be studied directly). This is about 500km in width and all light in the visible spectrum originates from this layer and has a temperature of around 5,500 °C.
- The Chromosphere, which is only observable during solar eclipses, is a dim red layer surrounding the photosphere. It also has a noticeable jump in temperature as it jumps to 25, 000 °C.
- The Corona, the outermost layer of the sun and also the hottest layer of the sun, and also the hottest layer of the sun, with a temperature of around 10^6 K.

Note also that the core, radiative zone and convection zone form the internal structure of the sun, while the photosphere, chromosphere and corona form the outer layer, or atmosphere of the sun [1].

A very prominent characteristic of the sun is its generation of sunspots. Sunspots are areas on the sun with incredibly large magnetic field (B) values, which in turn decreases the temperature of the sun within that area, and thus are darker in nature [6]. Sunspots typically consist of two regions, the umbra and the penumbra.

The umbra is the darker region focused in the centre of the sunspot, while the penumbra is the lighter region surrounding the internal umbra. The change

in colour from the photosphere to the sunspot is notably due to temperature differences from sunspot to the photosphere. While the photosphere has an average temperature of around 5,500 °C, the average sunspot temperature is about 3,400 °C.

Sunspots play a crucial role in this report, as they are the primary object that is observed on the photosphere to determine the differential rotation rate of the sun. Specifically when this report was written, a massive sunspot had developed in close proximity to solar equator. This proved to be an effective solar object to track the rotational movement across the sun.

2 Methodology

Most of this methodology will primarily be technologically/programming orientated. The equipment used within this report primarily consist of the North-West University's solar telescope, and a computer with the following applications:

- Helio-viewer
- GNU Image Manipulation Program (GIMP)
- Registax 6
- The Solar Ham website (<https://www.solarham.com/>)

2.1 Software

The process of determining the longitudinal differential rotation speed was as follows:

1. Using the solar telescope, centre the telescope onto the sun. Ensure that the skies are clear as clouds could obstruct the image.
2. Take a video of the sun using the telescope. For the purposes of this experiment, a video with 500 frames was used.
3. Repeat this step at the same hour on different days (ensuring that the weather allows such).
4. Once the videos have been taken and saved, load one of the videos into registax 6 to stack the frames atop one another, which will allow for a clearer and more informative image.
5. Once the image has been created and saved, load the image into GIMP, alongside the solarham image (taken on the same day). Add these two images as layers atop one another.
6. Within GIMP, vertically flip the image (to align it with the images taken in the northern hemisphere).

7. Then, with the solarham photo as a backdrop (that is, the solarham photo is in the layer behind the image taken with the solar telescope and the image has a slightly lower opacity), align the two photos so that their sunspots align.
8. Once the two images have been aligns, rotate the image slightly to account for differences in when the two images/videos were taken.
9. After this step, save the image and load it into Helio-viewer.
10. In Helio-viewer, once the image has been loaded, enable rotation-axis lines onto the image.
11. After this, click on the sunspot of interest to denote its latitude and longitude position.
12. Repeat the software steps for each video taken.

2.2 Method of data analysis

After the previous section, the data recorded is the latitude and longitude of the sunspot along the line. To then calculate the longitude differential speed, the following formula is used:

$$\text{Differential speed} = \frac{\text{longitudinal displacement (degrees)}}{\text{time passed (days)}} \quad (1)$$

This should then give the longitudinal or the latitudinal differential speed. In this report, the chosen sunspot only tracked movement over the longitudinal axis.

3 Results

For this results, the large sunspot near the centre of the solar disk was chosen and tracked over the course of measurements.

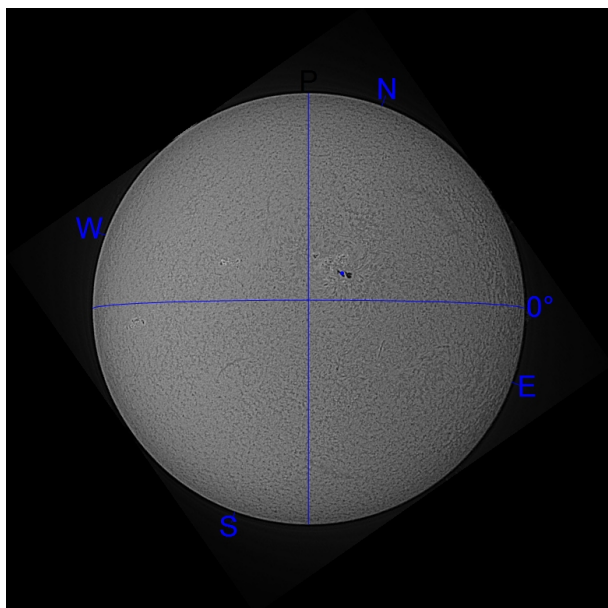


Figure 1: Helio-viewer rotational axis applied onto the 5th May solar telescope image

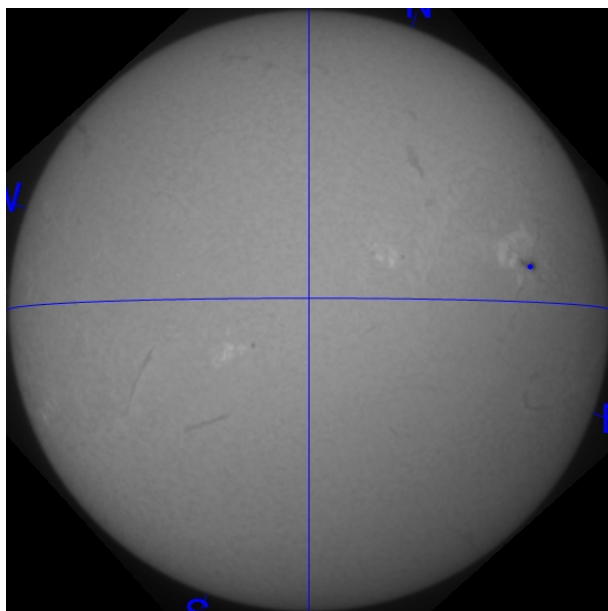


Figure 2: Helio-viewer rotational axis applied onto the 8th May solar telescope image

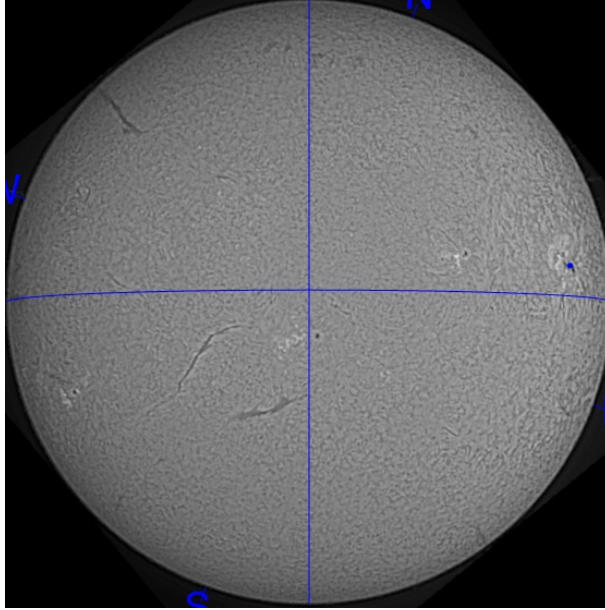


Figure 3: Helio-viewer rotational axis applied onto the 9th May solar telescope image

Date	Time of Measurement(Universal Time)	Latitude (degrees)	Longitude (degrees)
5th May	13:48	53	7
8th May	10:25	16	7
9th May	09:55	3	6

Table 1: Date, time and latitude measurement of chosen sunspot

Time between measurements (days)	Absolute difference in latitude (degrees)	Latitude per day (degrees/day)
3	37	12.33
1	13	13.00

Table 2: Calculation of latitude differential speeds

4 Discussion

The three images used in the report track a massive sunspot that is travelling closely to the solar equator. In the first image it is located near the solar prime meridian (where the longitude axis and the latitude axis intersect), but then over the following days, drifted towards the positive end of the latitude axis with minimal fluctuations across the longitudinal axis.

Comparing these images to solar images found on Helioviewer.org (under the image layer using observatory SDO with the instrument HMI), specifically the following images [4]:

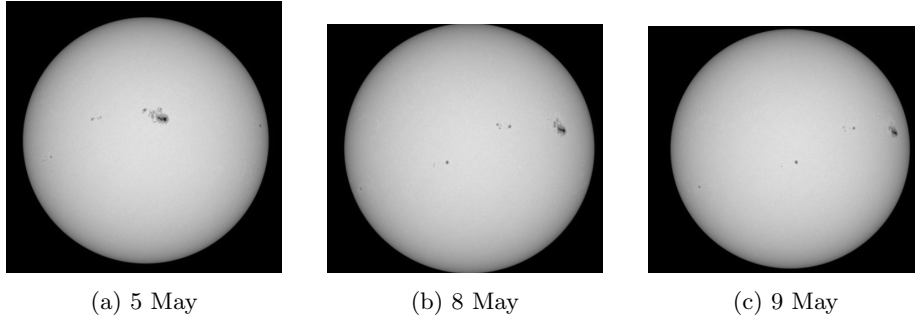


Figure 4: Helioviewer.org solar disk images over three days

These images are much less noisy than the images used within the report and that were edited using the programs listed in the methodology. Because of the increased clarity of these images, tracking the movement on the massive sunspot located near the equator could prove easier and hence give more accurate measurements.

Regarding the latitude differential speed, according to the Government of India's department of Science and Technology [2], the rotational speed of the sun near its equator is 13.98 degrees per day. The calculations performed here yielded a latitude differential speed of 12.33 degrees per day between day 1 and 2, and then a differential speed of 13.00 degrees per day between day 2 and 3.

5 Conclusion

The aim of this report was to determine the latitude differential rotation speed of the sun, using measurements from the NWU solar telescope in addition to software programs (Registax 6, Helio-viewer and GIMP). Measurements were taken across a near week long time frame, with measurements on the 5th of May, then 8th and 9th of May.

For the latitude differential rotation speed, the calculated speed, while close ($\pm$

1.50 degrees per day), still deviates a small amount from more accurate measurements. These accurate measurements measure the speed near the equator at about 14.00 degrees per day. This deviation could perhaps be attributed to a non-refined or careless use of the programs or perhaps an inconsistency to where the "dots" were placed in the Helio-viewer program. Latitude and longitude values can vary greatly by even a few millimetres difference in placement of the "dots". Overall, the result is still quite close. Perhaps more videos taken in days between, where the sunspot was still close the the solar prime meridian, would have yielded more accurate and consistent results.

As for the images, those taken and manipulated by the programs listed above and those collected from the Helioviewer.org website (under the observatory SDO), they contrast strongly because of the difference in "noise" between the SDO images and the images edited here, which could have lead to a difference in measurements. Even still, the original images still prove quite effective at tracking the latitude differential movement speed of the sunspot over the course of the week.

Overall, the report did come close to calculating the latitude differential speed of the sun near the solar equator, which implies great effectiveness of the NWU solar telescope as well as the implementation of these software that are all free. Another important fact is that with sites such as SolarHam and Helioviewer.org, and these free software then many individuals without access to high end equipment could still contribute greatly to the Solar Physics community as a whole leading to greater chances of important discoveries.

6 References

References

- [1] BYJU'S. Layers of the sun, 2022. Accessed: 2025-05-17.
- [2] Department of Science and Technology, India. Astronomers map differential rotation of sun's chromosphere using 100 years of kodaikanal data, 2022. Accessed: 2025-05-18.
- [3] Library of Congress. What is the transit of venus?, n.d. Accessed: 2025-05-17.
- [4] NASA Helioviewer Project. Helioviewer. <https://gs671-suske.ndc.nasa.gov/>, 2025. Accessed: 2025-05-18.
- [5] NASA Science. Our sun: Facts. <https://science.nasa.gov/sun/facts/>, 2020. Accessed: 2025-05-17.
- [6] National Weather Service. The sun and sunspots. <https://www.weather.gov/fsd/sunspots>, 2025. Accessed: 2025-05-17.